DeltaMask: Minimizing Communication in Federated Fine-Tuning of Vision Foundation Models

Vasileios Tsouvalas · Yuki M. Asano · Aaqib Saeed

Eindhoven University of Technology, The Netherlands | University of Technology Nuremberg, Germany

IEEE BigData 2024

01 Context & approach

The challenge

Federated adaptation of large vision models is expensive even with one-bit updates. *DeltaMask* keeps pretrained weights fixed and communicates selected changes in learned binary masks, reducing the information sent each round.

Our contribution

Adapt frozen vision models through stochastic binary masks.
Select informative mask changes and compress their indices with probabilistic filters.
Measure accuracy and bitrate across eight datasets and multiple pretrained architectures.

1 Learn masks

After one linear-probing round, freeze the backbone and optimize mask probabilities on each client.

2 Select and compress changes

Compare local and shared-seed server masks. Rank changed positions by KL divergence, then encode selected indices with Binary Fuse filters and DEFLATE.

3 Reconstruct and aggregate

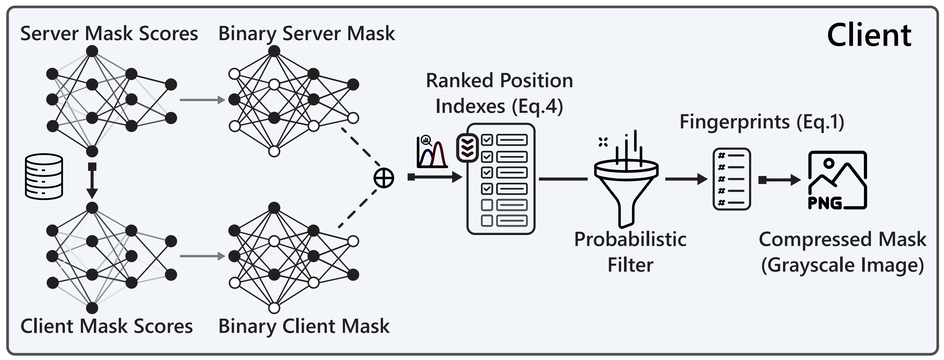
The server queries filter membership, flips the indicated mask bits, and updates global probabilities with Bayesian aggregation.

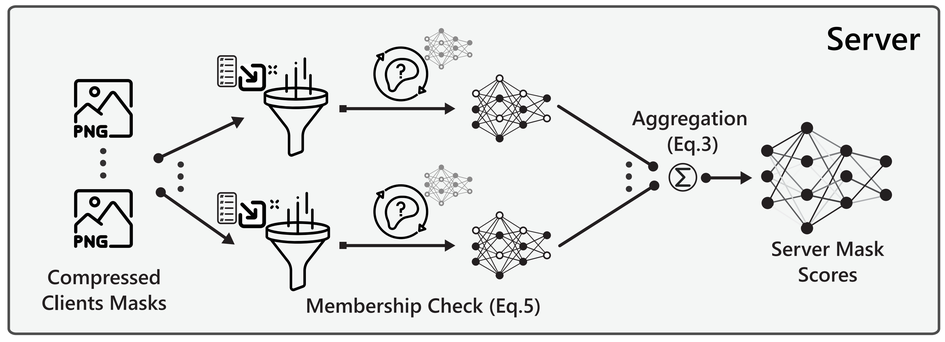
Scope

Probabilistic filters can add false-positive mask changes; accuracy remains below unrestricted fine-tuning. Evaluation covers image classification; broader task transfer is untested.

02 Method

Transmit compressed changes in stochastic masks instead of full pretrained weight updates.





Core formulation

$$\widetilde{w} = m \odot w_{\text{pre}}, \qquad m_i \sim \text{Bernoulli}(\theta_i)$$

w_{pre}: frozen pretrained weights; θ: learned keep probabilities; m: sampled binary mask. Adapt which weights are active.

$$\Delta = \{i : m_i^{\text{local}} \neq m_i^{\text{server}}\}, \qquad \Delta' = \text{TopK}_{\text{KL}}(\Delta)$$

Δ: changed mask indices; K: retained count. Rank changes by local-to-server Bernoulli KL divergence, then send selected indices.

Paper Code

03 Results

0.151 bpp
Average transmitted bitrate
8 datasets · CLIP ViT-B/32 · IID

85.44%
Average accuracy
Same setting · full participation

Low bitrate with a measured accuracy trade-off

Method	Avg. accuracy (%)	Avg. bits/parameter
Fine-tuning	88.08	32
FedMask	79.23	1.0
EDEN	84.14	0.691
FedPM	86.07	0.872
DeltaMask (Ours)	**85.44**	**0.151**

CLIP ViT-B/32 · 8 datasets · IID Dir(10) · 30 clients · full participation · 100 rounds. Means of 3 runs.

Non-IID data with 20% participation

Method	Avg. accuracy (%)	Avg. bits/parameter
Fine-tuning	83.19	32
EDEN	78.56	0.715
FedPM	81.64	0.948
DeltaMask (Ours)	**81.13**	**0.233**

Same 8 datasets and backbone · non-IID Dir(0.1) · 30 clients · 20% participation · 300 rounds. Means of 3 runs.

Takeaway

DeltaMask sends **selected mask changes**, reaching **0.151 bits/parameter** at **85.44% accuracy** across eight IID tasks. Compared with FedPM at 0.872 bits/parameter, it reduces communication while trailing accuracy by 0.63 percentage points.